What do you want to learn?

ALICE L WALTON SCHOOL OF MEDICINE

Mitral Stenosis

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Adapted from: Mitral Stenosis

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Learning Objectives (4) Versions

After completing this brick, you will be able to:

1. Define mitral stenosis and its clinical presentation.
2. Explain the etiology, pathophysiology, and complications of mitral stenosis.
3. Explain the diagnosis of mitral stenosis.
4. Describe the

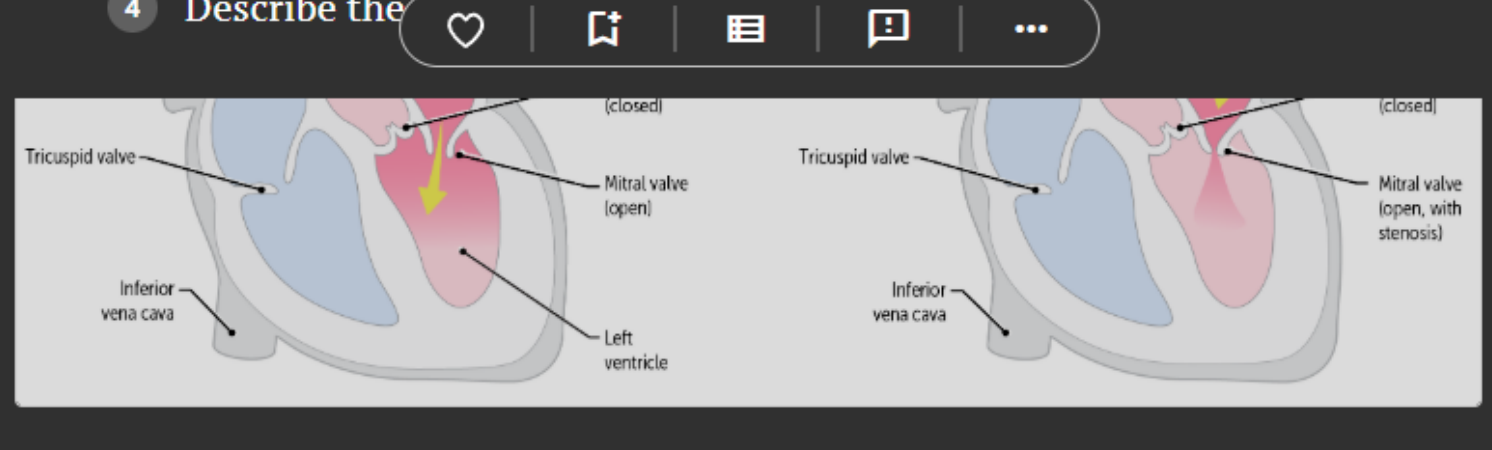
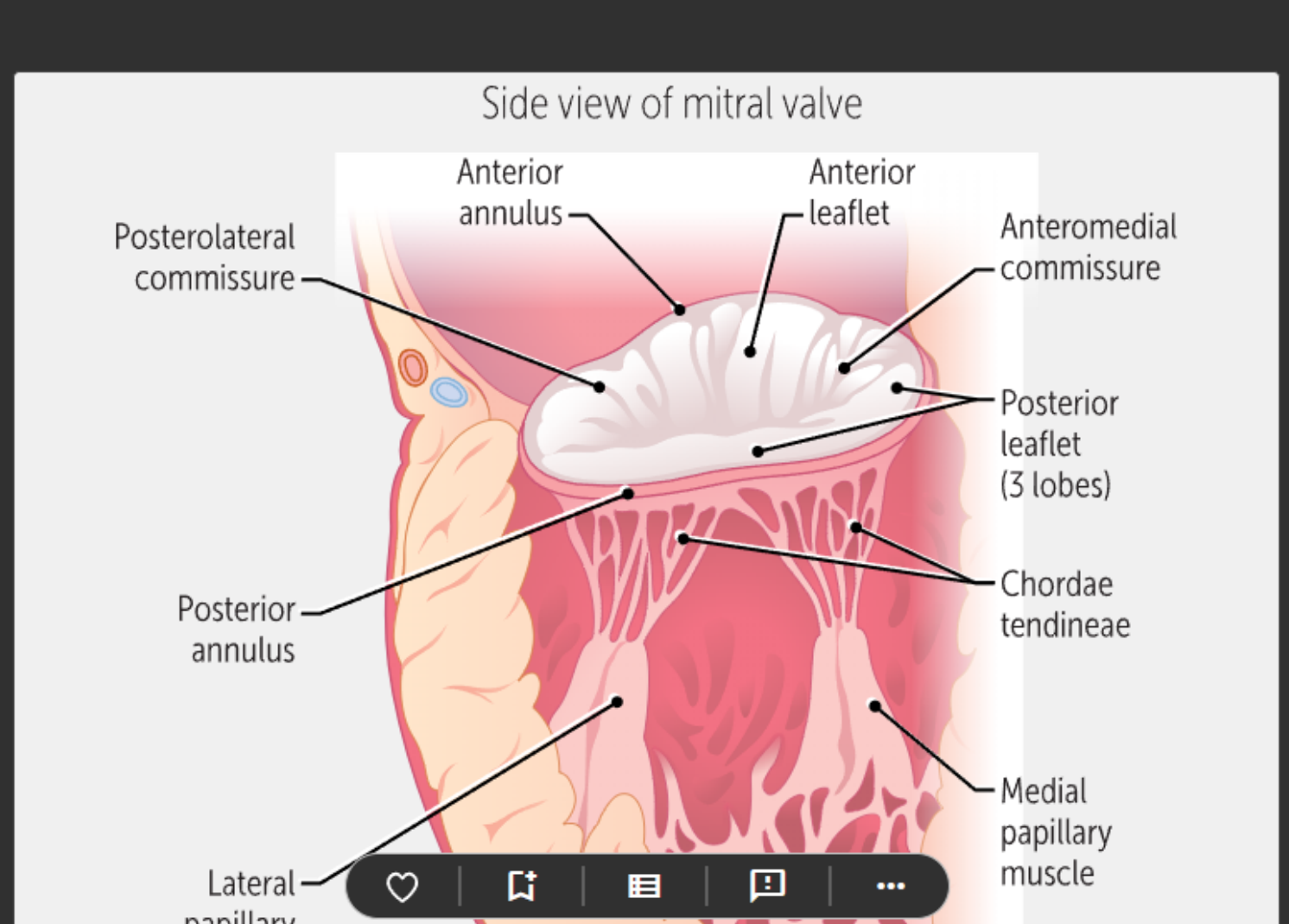


Figure 1 Mitral stenosis anatomy

Excess volume and pressure build up in the LA, while forward flow to the LV decreases. Let's review the components of the mitral valve (Figure 2).



may not always be aware of their symptoms, so taking a good history is important. Patients with either MS or Mitral Regurgitation may present with new onset atrial fibrillation caused by the atrial dilation (discussed below). The normal hemodynamic changes seen in pregnancy (increased heart rate and blood volume) may worsen symptoms or unmask previously asymptomatic MS (also discussed below).

Patients with MS may also experience hemoptysis. Increased pressure in the pulmonary circulation can cause capillaries in the lungs to rupture. Hemoptysis due to MS is less common now, due to earlier detection and management.

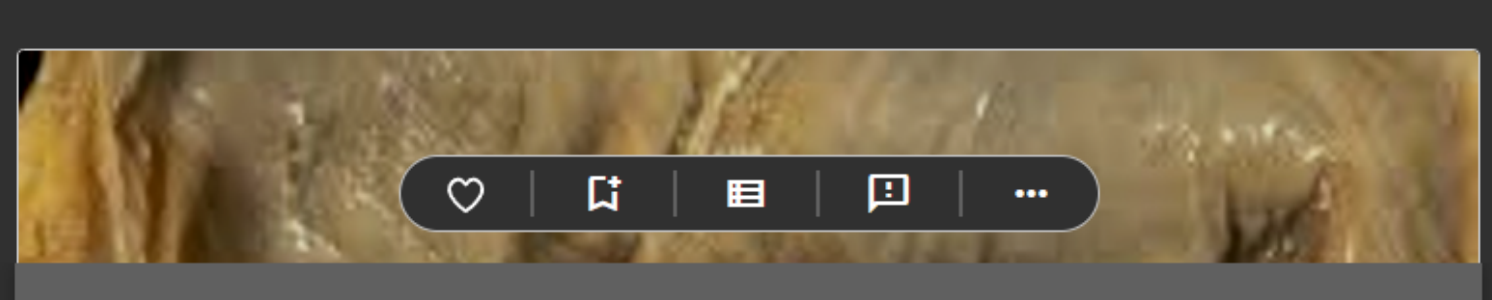
> CLINICAL CORRELATION
>
> Mitral Stenosis is more likely to cause hemoptysis than Mitral Regurgitation. Although both cause elevations in pulmonary vascular pressures which can lead to damage and ultimately bleeding from fragile pulmonary blood vessels, MS causes higher and more consistent elevations due to reduced flow across the mitral valve resulting in persistently greater left atrial pressures than MR.

What Is the Pathophysiology of Mitral Stenosis? (LO 2)

MS is caused by a variety of diseases that narrow the mitral valve. For comparison, a healthy cadaveric mitral valve is shown (Figure 3). Note the healthy **commissures** (ALC and PMC), the points where adjacent leaflets meet and join together. Th lve leaflet function, allowing them to separate and open/close properly. When these areas become

valve is the most common heart valve involved in RHD. Rheumatic fever is an inflammatory condition that can develop after inadequate treatment of streptococcal infections caused by group A streptococcus (GAS). It triggers an immune response to the antigens on GAS which are very similar to the structure of valvular tissue, resulting in autoimmune attack of the heart valves. Although the prevalence of rheumatic heart disease in developed nations like the United States is on the decline, an estimated 1 in 100,000 patients still contract this disease. Initially, rheumatic fever may present with Mitral Regurgitation. However, over a period of years to decades, chronic inflammation of the mitral valve leads to thickened, calcified leaflets of the valve with fibrin deposits on the cusps causing stenosis. Severe inflammation results in **commissural fusion** - the leaflets fuse together at the commissures. As a result, the stenotic mitral valve from rheumatic disease develops a characteristic slit-like, curved opening that resembles a fish mouth (Figure 4).

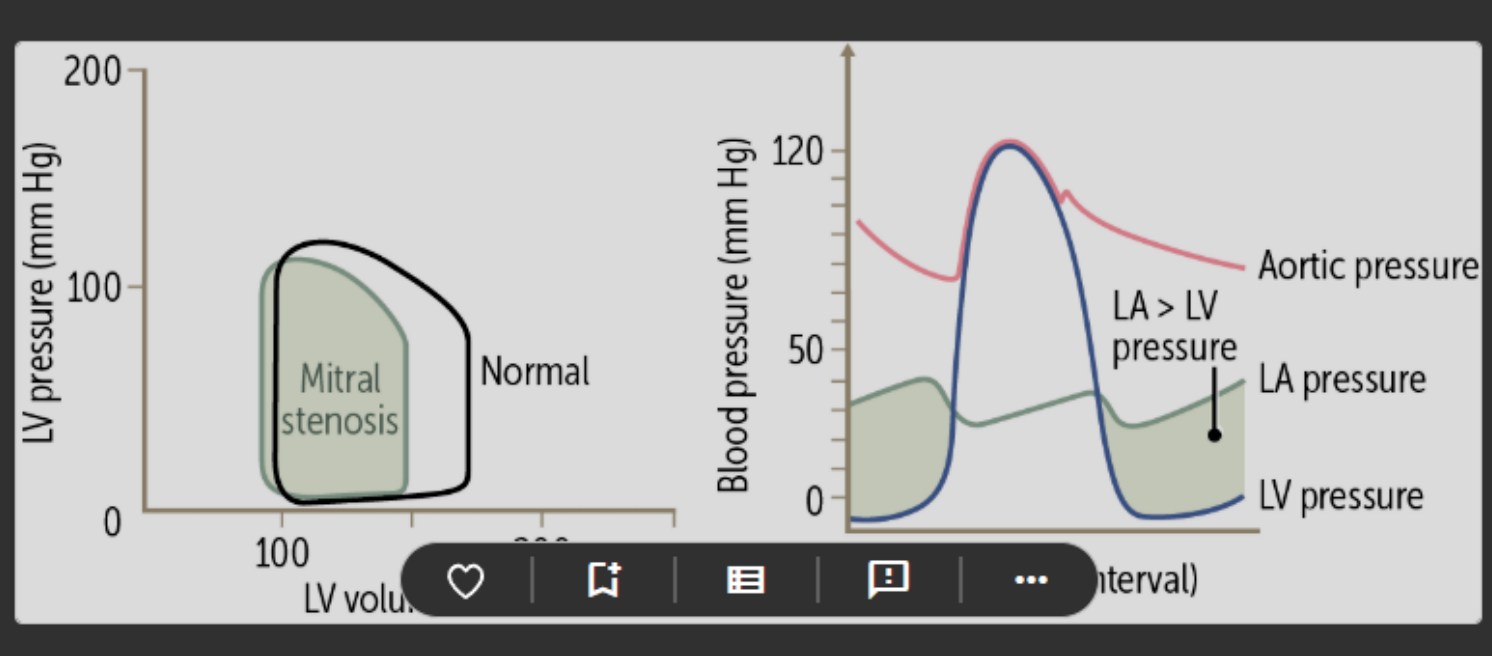
- **Calcification of Annulus.** In older patients calcium can deposit along the annulus. Annular calcification more commonly interferes with leaflet closure and results in Mitral Regurgitation. In advanced cases it can cause Mitral Stenosis by limiting the annulus's ability to expand during diastole and by restricting the movement of the valve leaflets (Figure 5). The mitral valve *annulus* is much less often calcified than the aortic valve, where calcification affects valve *leaflets*, because the aortic valve has a smaller diameter, higher rate of peak flow, and more turbulence.



What is the most common cause of Mitral Stenosis?

Pathophysiology of Mitral Stenosis (LO 2)

The main problem in MS is that the stenotic valve causes an increase in both LA volume and pressure during diastole (Figure 6, right).



> CLINICAL CORRELATION
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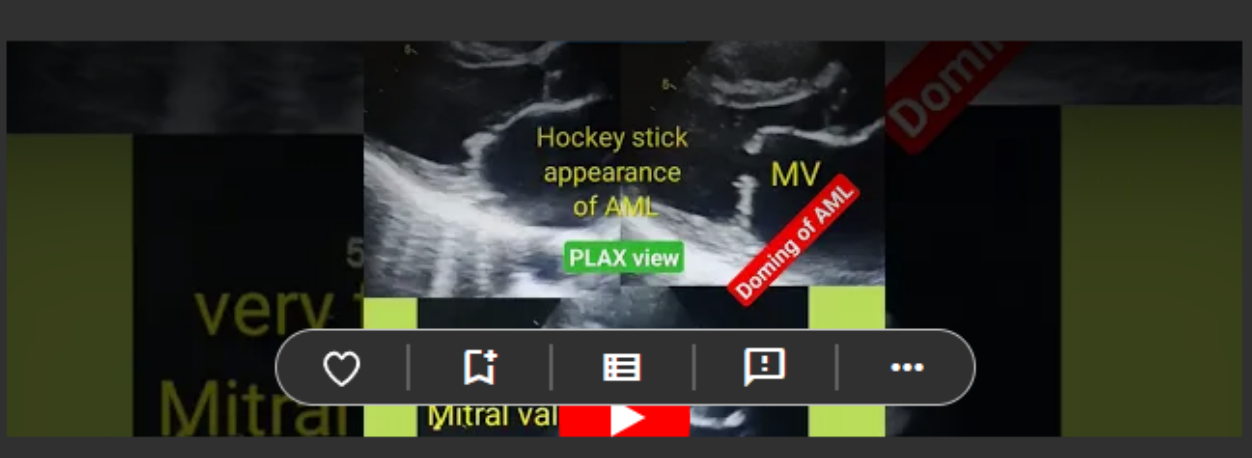
> Rheumatic **Mitral Stenosis** is the most common cardiac condition found in women during pregnancy. It is poorly tolerated due to the large increase in intravascular volume and increase in heart rate. Beta blockade can slow the heart rate and improve symptoms.
>
> Severe **Aortic Stenosis** is also poorly tolerated during pregnancy, since the cardiac output must increase significantly to perfuse the placenta and support the growing fetus. Increasing cardiac output in the setting of a narrowed Aortic Valve puts strain on the left ventricle.
>
> **Mitral Regurgitation** is better tolerated during pregnancy. The large increase in preload is offset by the decreased peripheral vascular resistance of pregnancy (estrogen and progesterone promote vasodilation, and the placental circulation is low-resistance). By lowering vascular resistance, the LV can eject more blood into the aorta and less blood back into the left atrium, decreasing the severity of the regurgitation.

Why do pregnant patients with MS tend to present symptomatically?

Why is an opening snap heard in mitral stenosis?

Imaging

Transthoracic echocardiogram (TTE) is indicated to establish a diagnosis by visualizing the stenotic valve orifice of Mitral Stenosis and to determine severity. A transesophageal echocardiogram (TEE) allows for better visualization of the LA and mitral valve as the ultrasound probe is inserted through the esophagus, which is nearer the LA. A TEE is a more invasive procedure, requiring sedation.



Medical Therapy

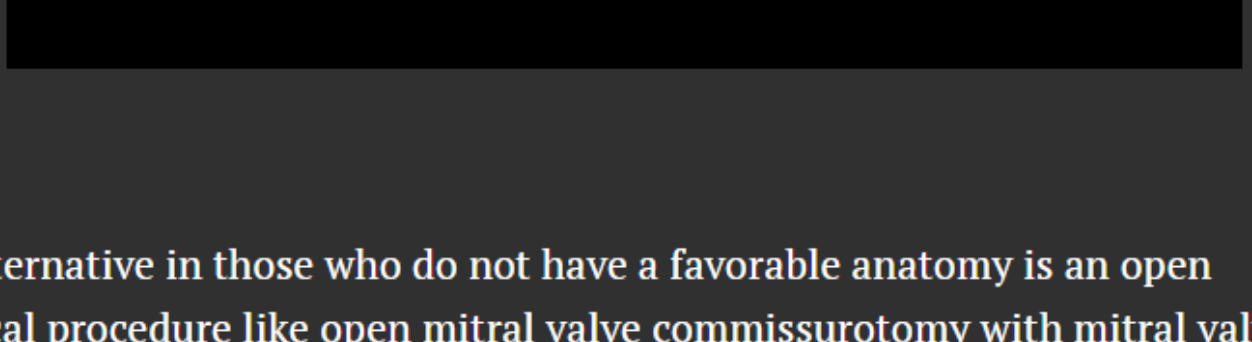
The first goal of medical treatment of MS is to control heart rate, typically with **β-blockers.** Why is this important? MS is an obstruction during diastole that inhibits passive filling of blood from the LA to the LV. As heart rate increases and the cardiac cycle shortens, nearly all of the time is subtracted from the diastolic phase. Slowing the heart rate allows more time for blood to flow through during diastole. Therefore, β-blockers work to reduce symptoms; however, they do not treat the underlying condition.

Because of the high risk of thromboembolism (>6% per year), patients with A fib and MS should take **warfarin** (an anticoagulant). The same applies to patients with MS and LA thrombus or a prior history of an embolism (eg, stroke).

Patients with volume overload (pulmonary edema) should be treated with **diuretics.**

Invasive Procedures

An invasive procedure is the only definitive treatment for MS. During **percutaneous mitral balloon commissurotomy** (aka valvotomy), a balloon tipped catheter is inserted via the femoral vein into the right atrium. A transeptal puncture then advanced the catheter to the left atrium and then across the mitral valve. The valve is dilated by a balloon (Figure 9). Percutaneous intervention is done in those with severe MS who have favorable valvular anatomy as determined by an echocardiographic scoring system.



An alternative in those who do not have a favorable anatomy is an open surgical procedure like open mitral valve commissurotomy with mitral valve repair. Here, the valve commissures are split and calcium deposits are removed. this requires a much longer recovery than the percutaneous approach, as does valve replacement.

Actual mitral valve replacement (using a mechanical valve or bioprosthetic valve of cow, pig, or human tissue) through open-heart surgery is now less common and is used for cases with extensive valve damage.

> CASE CONNECTION BACK TO INTRODUCTION ↑
>
> Thinking back to VM, how do you explain the cause of her condition? What do you tell her about treatment?
>
> You first reassure VM that her newborn son is doing very well, and she will see him soon. You tell her that although echocardiography is required for diagnosis, she appears to have mitral stenosis, a narrowing of one of her heart valves. This narrowing reduces blood flow across the valve and can cause fluid to build up in the lungs. You explain that MS is most often caused by rheumatic fever, an illness she likely had as a child. You reason that she developed pulmonary edema because the stenotic valve decreased the filling of the left ventricle during diastole and this is poorly tolerated with the large increase in blood volume during pregnancy. Also, th which further increases during the exertion of labor, decreases time for ventricular filling. You

Review Questions

1. Which of the following is not a cause of mitral stenosis?

A. ○ Congenital mitral stenosis
B. ○ Mitral annular calcification
C. ● Myocardial infarction
D. ○ Rheumatic heart disease

Hide Explanation

The correct answer is myocardial infarction (C). MI will not lead to stenosis of the mitral valve but may lead to mitral regurgitation due to papillary muscle rupture. Congenital Mitral Stenosis (A) is a condition in which patients are born with mitral stenosis (MS). Mitral annular calcification (B) occurs when the annulus, the ring-like structure holding the leaflets in place, becomes calcified and causes narrowing of the mitral valve. Rheumatic heart disease (D) is the most common cause of Mitral Stenosis and causes narrowing of the valve by inducing inflammation and scarring of the leaflets.

2. Which of the following is the most common symptom of mitral stenosis?

A. ○ Cough
C. ○ Diuretics
D. ○ Heart surgery
E. ● Percutaneous mitral balloon commissurotomy

Hide Explanation

The correct answer is Percutaneous Mitral Balloon Commissurotomy (E), which has a much faster recovery than surgical repair or replacement of the valve (D), which is reserved for more severe cases in which balloon commissurotomy is not an option. β-Blockers (A) and calcium channel blockers (B) can help control the heart rate, which may be elevated because atrial fibrillation or decreased forward flow, but they will not correct the valve itself. Diuretics (C) are medications that can help patients eliminate excess fluid, which can provide relief in patients with pulmonary edema. However, diuretics will not correct the problem with the mitral valve itself.

4. Which of the following describes the characteristic murmur of mitral stenosis?

A. ● Diastolic murmur with an opening snap, followed by a low-pitched diastolic rumble at the apex
B. ○ Harsh systolic crescendo-decrescendo murmur at the right sternal border, second intercostal space that radiates to the carotids
C. ○ High-pitched, blowing, holosystolic murmur at the left fifth intercostal space at the midclavicular line radiating to the axilla
D. ○ Soft, high ur at the left sternal border, third intercostal space,

diagnosis of all valvular disorders and assessing their severity. An electrocardiogram (A) is useful for evaluating patients with mitral stenosis for atrial fibrillation, as it measures the electrical activity of the heart. Magnetic resonance imaging (C) can be useful for evaluating the structure and anatomy of the heart but is a second-line diagnostic tool; echocardiography is still preferable. X-ray (D) provides a two-dimensional image of the heart and is less informative than echocardiography, but it may identify calcification of the mitral or aortic valves.

How would you rate this Brick?

References

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